

From Sum-Rate to Energy-Efficient Beamforming: What Is Reused, What Is Extended, and What Is Novel

1 Purpose of this note

This note separates three things that are easy to blur together when presenting the idea: (i) what is taken directly from the reference paper [1], (ii) what we changed, and (iii) why the change is a genuine departure and not a relabeling of the same optimum. The system and channel model in Section 2 is unchanged from the paper and is restated only to fix notation. Section 7 positions this extension against the closest related work found outside the reference paper’s own line of work.

2 System model (adopted, unchanged)

A set $\mathcal{M} = \{1, \dots, M\}$ of LEO satellites, each equipped with a uniform linear array of N antennas, serves a set $\mathcal{K} = \{1, \dots, K\}$ single-antenna users in the downlink. The channel between satellite m and user k is $\mathbf{h}_{k,m} \in \mathbb{C}^{1 \times N}$, stacked as $\mathbf{h}_k = [\mathbf{h}_{k,1}, \dots, \mathbf{h}_{k,M}] \in \mathbb{C}^{1 \times MN}$. Data symbols are precoded with $\mathbf{w}_k \in \mathbb{C}^{MN \times 1}$ and superimposed, $\mathbf{x} = \sum_{k \in \mathcal{K}} \mathbf{w}_k u_k$. The full precoding matrix is $\mathbf{W} = [\mathbf{w}_1, \dots, \mathbf{w}_K] \in \mathbb{C}^{MN \times K}$. The instantaneous achievable sum rate is

$$R(\mathbf{W}) = \sum_{k \in \mathcal{K}} \log_2 \left(1 + \frac{|\mathbf{h}_k \mathbf{w}_k|^2}{\sigma_{\text{noise}}^2 + \sum_{k' \neq k} |\mathbf{h}_k \mathbf{w}_{k'}|^2} \right). \quad (1)$$

CSIT is corrupted by two error sources acting on the true channel $\mathbf{h}_{k,m}$: an angle-of-departure (steering) error $\epsilon_{\text{aod},k,m} \sim \mathcal{U}(-\Delta\epsilon_{\text{aod}}, \Delta\epsilon_{\text{aod}})$ and a synchronization phase error $\epsilon_{\text{ph},k,m} \sim \mathcal{N}(0, \sigma_{\text{ph}}^2)$, giving an estimate $\tilde{\mathbf{h}}_{k,m}$ used by every precoder in place of the true $\mathbf{h}_{k,m}$.

3 The paper’s optimization problem (reproduced)

The reference paper poses a single objective: maximize expected sum rate under a per-satellite radiated-power budget,

$$\max_{\mathbf{W}} \bar{R} = \mathbb{E}_{\mathbf{H}}[R(\mathbf{W})] \quad \text{s.t.} \quad \|\mathbf{W}_m\|_F^2 \leq P/M \quad \forall m \in \mathcal{M}. \quad (2)$$

Every precoder evaluated in the paper — MMSE, robust SLNR, and the learned SAC policy — enforces this constraint by *unconditional rescaling* of the raw precoder to the boundary of the feasible ball:

$$\mathbf{W}_{\text{paper}} = \sqrt{\frac{P/M}{\|\mathbf{W}_{\text{raw}}\|_F^2}} \mathbf{W}_{\text{raw}}, \quad (3)$$

independent of $\|\mathbf{W}_{\text{raw}}\|_F^2$. This is Eq. (26)/(30) of the paper for MMSE, and the SAC output transform is normalized identically. The reward signal driving SAC’s actor is the sum rate $R(\mathbf{W}_{\text{paper}})$ alone; power never appears as a cost, only as this hard, always-active rescaling.

Lemma 1 (Full-power optimality under rate-only objectives). *Fix a precoding direction $\hat{\mathbf{W}}$ with $\|\hat{\mathbf{W}}\|_F^2 = 1$ and consider the scaled family $\mathbf{W}(\alpha) = \alpha \hat{\mathbf{W}}$, $\alpha \in (0, \sqrt{P/M}]$. Then $R(\mathbf{W}(\alpha))$ is non-decreasing in α , so the maximizer of (2) over this family is $\alpha^* = \sqrt{P/M}$.*

Sketch. Scaling $\mathbf{W}$ by α scales every signal term $|\mathbf{h}_k \mathbf{w}_k|^2$ and every interference term $|\mathbf{h}_k \mathbf{w}_{k'}|^2$ by α^2 , while σ_{noise}^2 is fixed. Hence each user’s SINR, $\alpha^2 |\mathbf{h}_k \hat{\mathbf{w}}_k|^2 / (\sigma_{\text{noise}}^2 + \alpha^2 \sum_{k' \neq k} |\mathbf{h}_k \hat{\mathbf{w}}_{k'}|^2)$, is non-decreasing in α (strictly increasing while noise-limited, saturating to an interference-limited ceiling as $\alpha \rightarrow \infty$, never decreasing). $\log_2(1 + \cdot)$ is monotone, so R inherits the same monotonicity. ■ □

Lemma 1 is not a limitation of the paper’s method — it is the *correct* optimal behavior for problem (2). Rescaling every raw output to the boundary (Eq. 3) is therefore not a modeling simplification; it exactly implements the optimal solution structure of a rate-only objective. Any precoder solving (2) *should* saturate the power budget.

4 Our extension: making power a cost, not just a cap

4.1 Power consumption model (new)

The paper’s constraint counts only radiated power, $\|\mathbf{W}\|_F^2$. We add a satellite HPA electrical power model of the same form used in recent energy-efficient satellite precoding work [14], splitting DC power draw into amplifier draw and static circuit dissipation:

$$P_{\text{total}}(\mathbf{W}) = \underbrace{\frac{\text{tr}\{\mathbf{W}^H \mathbf{W}\}}{\eta_{\text{PA}}}}_{\text{amplifier draw}} + \underbrace{MN P_c}_{\text{circuit draw}}, \quad (4)$$

with power-amplifier efficiency $\eta_{\text{PA}} \in (0, 1)$ and per-antenna static circuit power P_c . We set $\eta_{\text{PA}} = 0.60$, matching the satellite antenna/HPA efficiency used in [14] and consistent with reported saturated TWTA/SSPA efficiencies of 55–70% [15]. An earlier version of this model used $\eta_{\text{PA}} = 0.35$ sourced from a *terrestrial* macro/micro/pico/femto base-station power study [5] – not satellite hardware – and has been corrected. Two structural properties of (4) matter for what follows:

- **No such term exists in the reference paper.** Its optimization problem (2) has no notion of amplifier efficiency or circuit power; P_{total} is our addition in full.
- P_{total} has a strictly positive floor $MNP_c/\eta_{\text{PA}} \cdot 0 + MNP_c = MNP_c$ as $\text{tr}\{\mathbf{W}^H \mathbf{W}\} \rightarrow 0$: the circuit term never vanishes. This bounds the maximum achievable percentage power savings and keeps any rate/power ratio well defined even at zero transmit power.

4.2 The energy-efficiency objective (new)

We replace (2) with a fractional (concave-over-affine) objective,

$$\max_{\mathbf{W}} \text{EE}(\mathbf{W}) = \frac{R(\mathbf{W})}{P_{\text{total}}(\mathbf{W})} \quad \text{s.t.} \quad \|\mathbf{W}_m\|_F^2 \leq P/M \quad \forall m \in \mathcal{M}. \quad (5)$$

This is a genuinely different problem from (2), not a relaxation of it: power now appears in the objective, not only in the constraint. Lemma 1 no longer applies, because scaling $\mathbf{W}(\alpha) = \alpha \hat{\mathbf{W}}$ now moves *both* the numerator and the denominator:

$$\text{EE}(\alpha) = \frac{R(\alpha \hat{\mathbf{W}})}{\alpha^2 \|\hat{\mathbf{W}}\|_F^2 / \eta_{\text{PA}} + MNP_c}. \quad (6)$$

The numerator saturates (Lemma 1, bounded above by the interference-limited ceiling); the denominator grows without bound, linearly in α^2 . Consequently $\text{EE}(\alpha)$ is not monotone: it generically increases while noise-limited, then *decreases* once the marginal rate gained per additional watt falls below the marginal cost of that watt. The maximizer α^* of (6) is generically in the *interior* of $(0, \sqrt{P/M}]$, not at the boundary. This is the precise sense in which the idea is not a relabeling of the paper’s problem: we did not merely permit $\|\mathbf{W}\|_F^2 < P/M$ (that permission alone is vacuous under (2), by Lemma 1); we changed the objective so the boundary stops being automatically optimal.

Remark 1. *This is precisely the classical energy-efficiency/spectral-efficiency trade-off [6]: under a rate-only objective, more power is always weakly better and the power cap binds; under a rate-per-power objective, the optimal power level is an interior stationary point of a concave-over-affine ratio, characterized by Dinkelbach’s transform (Section 5).*

4.3 Relaxing the always-saturate precoder (new)

To let the learned policy actually reach the interior of the feasible ball, we replace the unconditional rescaling (3) with a *clip-only* projection,

$$\mathbf{W}_{\text{ours}} = \begin{cases} \mathbf{W}_{\text{raw}}, & \|\mathbf{W}_{\text{raw}}\|_F^2 \leq P/M, \\ \sqrt{P/M} / \|\mathbf{W}_{\text{raw}}\|_F \cdot \mathbf{W}_{\text{raw}}, & \text{otherwise.} \end{cases} \quad (7)$$

Constraint (15b) of the paper is enforced exactly as written (an inequality) rather than as the de facto equality every paper precoder produces. This is the actual, literal feasible set of (5); (3) only ever explores its boundary.

5 Three formulations implemented and compared

Scheme I — reproduction of the paper’s convention. Precoder (3), reward $R(\mathbf{W})/P_{\text{total}}$. Since P_{total} is a constant here (power is always exactly P/M), this reduces to maximizing $R(\mathbf{W})$ alone — it is the paper’s sum-rate objective in all but name, retained purely as a same-conditions reference point.

Scheme II — direct-ratio reward, free power. Precoder (7), reward $R(\mathbf{W})/P_{\text{total}}(\mathbf{W})$ evaluated directly as a ratio. Numerically fragile as $P_{\text{total}} \rightarrow MNP_c$ from below is bounded (Section 4), but the ratio’s gradient can still be poorly scaled for online SGD.

Scheme III — adaptively-priced reward, free power. Precoder (7); reward uses the algebraic form of the Dinkelbach transform [7] — $R(\mathbf{W}) - \lambda P_{\text{total}}(\mathbf{W})$ turns a ratio objective into a subtractive one, which is what makes it usable as an RL reward at all — but not the Dinkelbach *algorithm*:

$$\text{reward}_t = R(\mathbf{W}_t) - \lambda_t P_{\text{total}}(\mathbf{W}_t), \quad \lambda_{t+1} = \frac{\bar{R}_{\text{window}}}{\bar{P}_{\text{total,window}}}, \quad (8)$$

where λ_t is a running EMA of realized rate/power, updated every transition, concurrently with a policy that is never frozen for any inner solve. The classical scheme alternates: solve $F(\lambda) = \max_{\mathbf{W}}[R(\mathbf{W}) - \lambda P_{\text{total}}(\mathbf{W})]$ to exact global optimality for a *fixed* λ , *then* update λ from $F(\lambda^*) = 0 \iff \lambda^* = \text{EE}(\mathbf{W}^*)$. Neither half of that is what (8) does. This is not a relaxation of Dinkelbach’s method in the sense of an inexact version of the same algorithm — it is a structurally different object (a continuously co-adapting price and policy) that borrows the transform’s algebra without either the exact inner solve or the alternation. Section 6 treats it on those terms.

6 What can and cannot currently be said about the online update

An earlier version of this section tried to show that (8) is a rigorously-justified *inexact instance* of Dinkelbach’s method, via a three-condition inexact-fractional-programming theorem from Jin et al. [20]. Re-reading that paper in full, and re-checking the codebase and the other cited results directly rather than relying on how they are usually summarized, that argument does not hold up. This section replaces it with what is actually true: two structural facts that hold regardless of what the mechanism is called (Sections 6.2–6.3), one honest but weaker positive result (Section 6.4), empirical evidence bearing on — but not proving — whether training settles at all (Section 6.5), and a direct statement of what remains open (Section 6.6).

6.1 Why (8) is not Dinkelbach’s method, even inexactly

Dinkelbach’s method alternates: solve $F(\lambda) = \max_{\mathbf{W}}[R(\mathbf{W}) - \lambda P_{\text{total}}(\mathbf{W})]$ to exact global optimality for a *fixed* λ , then update λ from that optimum. (8)’s active update path does neither half: λ updates every transition, unconditionally, via an EMA with fixed window (constant blend factor $2/(5000+1) \approx 4 \times 10^{-4}$, never decayed), and the policy is never frozen for any inner solve at any λ . This is a continuously co-adapting price-and-policy system that borrows the transform’s algebra, not the algorithm.

Checking whether Jin et al. [20] nonetheless covers this: it does not. Their linear-convergence theorem (their Theorem 1) is proven exclusively for a *tabular* Fractional Q-Learning algorithm over a finite state-action space, whose shrinking inner-loop error bound is enforced by a tabular sub-routine (Speedy Q-Learning, a known-convergent tabular method) with its own established convergence rate. The paper separately proposes a deep-RL extension (D3QN + DDPG for their hybrid discrete/continuous action space) for its experiments — structurally the closest thing to SAC anywhere in the paper — but states no convergence theorem for it; that extension is validated only empirically, via a results figure, and the paper itself notes the resulting convergence curve “may sometimes change non-monotonically.” Mapping SAC onto the paper’s three conditions targets a theorem that was never proven for a continuous-action, neural-function-approximated policy — the paper leaves that case exactly as open as this document does.

Two further citations require correction. This document previously justified skipping the current primal-dual RL literature [18, 19] on the grounds that it “only” applies to a softmax policy over a finite action space. Read directly, neither paper says that. Paternain et al. [18] formulate their constrained-RL setting over compact *continuous* action spaces from the outset, and separately prove their zero-duality-gap result extends to neural-network policy parametrizations, with optimality loss “made arbitrarily small” for near-universal approximators. The arXiv identifier

previously cited for Ding, Zhang, and Başar (2020) in fact resolves to a later, five-author journal extension [19] of that conference paper; that extended paper’s strongest, dimension-free rate is for softmax specifically, but it separately establishes sublinear convergence for “general smooth policy parametrizations” — which would include neural policies — up to a function-approximation-error term. Neither paper is a drop-in fit here (both are constrained RL, with a Lagrange multiplier enforcing a hard constraint, not a Dinkelbach-style ratio objective), but the stated reason for setting them aside was incorrect, and on reflection they look like a more promising home for a future rigor pass than the tabular result this section leaned on instead.

Finally, the two citations offered as “an explicit, checkable route” to a shrinking approximation-error bound, Ghadimi–Lan [21] and Ghadimi–Lan–Zhang [22], prove convergence of projected stochastic gradient methods to a *stationary point* of a *fixed, time-invariant* objective. Neither condition transfers: stationarity does not imply near-global-optimality for a nonconvex objective, and $f_\lambda(\mathbf{W}) = R(\mathbf{W}) - \lambda P_{\text{total}}(\mathbf{W})$ genuinely is nonconvex in $\mathbf{W}$ — `calc_sum_rate.py` computes exactly the standard interference-coupled SINR sum, the textbook source of MU-MIMO sum-rate’s non-concavity, and Lemma 3 below proves only smoothness, never convexity, in $\mathbf{W}$. Both theorems also require λ held fixed while the optimization runs, which the active EMA never does.

Ha et al. [14], by contrast, is accurately characterized here: its Algorithm 1 genuinely alternates — solve the inner problem, update η , repeat to convergence — the textbook Dinkelbach structure, and a useful contrast case for what (8) is not. Deng et al. [16] is a real, correctly-attributed paper; its precise inner-solve mechanism was not independently re-verified for this revision and should be treated as unconfirmed rather than as supporting evidence.

6.2 What holds regardless of the mechanism’s name

Lemma 2 (Action-independent transitions collapse temporal credit assignment). *Let s_t denote the SAC state at training step t and a_t the (pre-projection) action. If the state-transition kernel satisfies $s_{t+1} \sim p(\cdot \mid s_t)$, independent of a_t , then for every state s ,*

$$Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim p(\cdot \mid s)}[V^*(s')], \quad \arg \max_a Q^*(s, a) = \arg \max_a r(s, a).$$

Proof. By the Bellman optimality decomposition, $Q^*(s, a) = \mathbb{E}[r_t + \gamma V^*(s_{t+1}) \mid s_t = s, a_t = a]$. Since $s_{t+1} \sim p(\cdot \mid s_t)$ does not depend on a_t , the second term equals $\gamma \mathbb{E}_{s' \sim p(\cdot \mid s)}[V^*(s')]$ for every a — a quantity constant in a . Adding an a -independent term to $r(s, a)$ does not change its argmax. ■

Remark 2 (This holds by construction in this codebase, not by assumption). `update_sim()`, the state-transition routine shared by every training script in this repository, advances user/satellite positions, rolls CSIT estimation errors, and recomputes channel state information without ever receiving the chosen precoder as an argument. The hypothesis of Lemma 2 is therefore exact, not approximate: whichever $\mathbf{W}_t$ SAC outputs, the channel realization at step $t + 1$ is drawn from the same exogenous process regardless. There is no delayed consequence of a precoding decision anywhere in this environment — it is a per-sample stochastic optimization problem wearing an RL interface, not a genuine sequential MDP.

A second, independent fact makes this stronger than the environment’s structure alone requires: `config_sac_learner.py` sets `future_reward_discount_gamma` = 0, and `train_graph` gates its entire next-state-bootstrapping branch behind $\gamma > 0$. With $\gamma = 0$ that branch never executes — the critic’s regression target is literally `target_q = rewards`, not a bootstrapped return. The critic is not learning a discounted value function that Lemma 2 happens to simplify; by this configuration

choice alone, and regardless of whether the environment is action-independent, it is directly regressing immediate reward. Lemma 2 explains why that choice costs nothing relative to the general (bootstrapped) case; it is not needed to establish what the trained critic’s target actually is.

Remark 3 (What this does and does not license). *Lemma 2 removes temporal credit assignment as a concern. It says nothing about whether a concurrently-updated λ converges to anything in particular, nor about whether SAC’s inner optimization (for whatever λ it currently faces) reaches a good solution. Those questions are addressed, and left partly open, in Sections 6.4–6.6.*

6.3 Smoothness of the fixed- λ reward surface

For a fixed λ , $F(\lambda) = \max_{\mathbf{W} \in \mathcal{W}} [R(\mathbf{W}) - \lambda P_{\text{total}}(\mathbf{W})]$ — the value of the *exact*, globally-solved inner problem — is convex in λ on $\mathbb{R}$ (a pointwise supremum of functions affine in λ is convex, regardless of the form of R or P_{total}). This is a real property of the idealized optimization landscape and is worth recording, but it characterizes $F(\lambda)$ itself, not anything SAC computes: SAC never solves the inner problem to global optimality, so this convexity-in- λ fact does not, by itself, say anything about the policy the training procedure actually reaches. The property that does bear on the trained policy is smoothness *in $\mathbf{W}$* , for fixed λ :

Lemma 3 (Smoothness of the per-step Dinkelbach surface). *Fix $\lambda \geq 0$ and let $f_\lambda(\mathbf{W}) = R(\mathbf{W}) - \lambda P_{\text{total}}(\mathbf{W})$. On $\mathcal{W}$, f_λ is continuously differentiable with Lipschitz-continuous gradient (finite L_λ -smooth).*

Sketch. $\sigma_{\text{noise}}^2 > 0$ is a fixed positive constant, so every SINR denominator in (1) is bounded away from zero uniformly on $\mathcal{W}$; R is a composition of rational functions (bounded denominator) with $\log_2(1 + \cdot)$, hence real-analytic on $\mathcal{W}$. P_{total} (4) is affine in the quadratic form $\text{tr}\{\mathbf{W}^H \mathbf{W}\}$, with constant Hessian $(2/\eta_{\text{PA}})I$. A finite linear combination of functions with continuous Hessian has bounded (hence Lipschitz) Hessian on the compact set $\mathcal{W}$. ■ □

Remark 4 (The clip is a projection, not an objective-level discontinuity). *The precoder (7) is exactly Euclidean projection $\Pi_{\mathcal{W}}$ onto a convex ball. The composite map $\mathbf{W}_{\text{raw}} \mapsto f_\lambda(\Pi_{\mathcal{W}}(\mathbf{W}_{\text{raw}}))$, viewed as a function of the unconstrained actor output, has a discontinuous Jacobian exactly on the sphere $\|\mathbf{W}_{\text{raw}}\|_F^2 = P/M$ — $\Pi_{\mathcal{W}}$ itself is what is non-smooth there, not f_λ . This is the setting nonconvex projected-stochastic-gradient theory is built for when λ is fixed: convergence there is stated via the gradient mapping $G_\lambda(\mathbf{W}) = (\mathbf{W} - \Pi_{\mathcal{W}}(\mathbf{W} + \alpha \nabla f_\lambda(\mathbf{W}))) / \alpha$ rather than the raw gradient [22], sidestepping the projection boundary’s non-smoothness instead of requiring it away. Lemma 3 is the hypothesis on f_λ that framework needs — but the framework itself, per [21, 22], additionally requires λ to be held fixed while it runs, and the active per-step EMA in (8) never freezes λ . Lemma 3 is therefore evidence that gradient-based training is locally well-behaved at any instant, not a license to invoke this convergence framework as stated.*

6.4 A weak, honest positive result: self-consistency, not optimality

One argument can be made without borrowing convergence machinery that does not apply. Suppose the joint process (λ_t, θ_t) (θ the policy parameters) reaches a stationary point: λ stops moving, the policy stops changing. Two things are then true *by construction*, not by assumption: λ must equal the realized $R(\theta)/P_{\text{total}}(\theta)$ exactly, since λ is defined as an EMA tracking that ratio and an EMA of a constant sequence converges to that constant; and the policy must be a (soft) maximizer of $R - \lambda P_{\text{total}}$, since that is what SAC’s training objective pushes toward for a stationary reward. Together these are exactly Dinkelbach’s fixed-point condition, $\lambda^* = R(\mathbf{W}^*)/P_{\text{total}}(\mathbf{W}^*)$ at a maximizer of

$R - \lambda^* P_{\text{total}}$ — so if the dynamics settle anywhere, they settle somewhere self-consistent with the Dinkelbach optimality condition.

Remark 5 (What this does not show). *This is a characterization of self-consistency, not of optimality, and it is important not to overstate it. Dinkelbach’s classical equivalence between $F(\lambda^*) = 0$ and global ratio-optimality relies on $\mathbf{W}^*$ being the global maximizer of $R - \lambda^* P_{\text{total}}$; here θ is at best a local stationary point of a nonconvex training objective. A policy that transmits at the circuit-power floor with near-zero rate is, in this weak sense, just as much a “fixed point” as a good one — $\lambda \rightarrow 0$ and $R - \lambda P_{\text{total}} \approx 0$ are self-consistent too. This argument establishes that λ correctly describes whatever the policy ends up doing; it says nothing about why the policy ends up doing something good, which is a separate, much harder claim this document does not make. It is also unclear the premise is ever exactly satisfied: both optimizer learning rates in `config_sac_learner.py` are constant (no decay schedule is active), and constant-step-size stochastic optimization generically settles into a noisy neighborhood rather than a literal fixed point.*

6.5 Empirical evidence, honestly bounded

Two checks against an active training run (job with the current, reverted `EE.sac.py`, plain `lwin5000` mode) bear on, without proving, whether the dynamics in Section 6.4 settle in practice.

Timescale separation is real but not asymptotic. Episode-to-episode, λ ’s mean relative change is consistently about $4\times$ smaller than that of the realized R/P_{total} ratio it tracks, stable across three non-overlapping windows spanning roughly 4000 episodes (ratios 4.03, 3.91, 4.03). This is genuine evidence of practical separation — not asserted from the EMA window size, measured from training logs — but it is a constant ratio, not one diverging toward the $a(n)/b(n) \rightarrow 0$ condition classical two-timescale stochastic-approximation theory needs. λ ’s and the realized ratio’s episode means also track each other almost exactly throughout (e.g. 2.679 vs. 2.682 early, 3.299 vs. 3.300 late), which is closer to λ being a smoothed shadow of one signal than to two variables settling into separately-decoupled equilibria.

The entropy-driven bias described below is not shrinking, and by one measure is still growing. α_{SAC} (the auto-tuned entropy temperature) rises monotonically across the full range checked so far (episode-window means $0.32 \rightarrow 7.77 \rightarrow 8.85 \rightarrow 9.20$ over roughly 5100 of 13,000 episodes), with shrinking but nonzero variance — not the plateau a first glance at a handful of adjacent episodes suggests. This is consistent with, and does not contradict, the entropy-bias diagnosis below; if anything it shows the bias is still actively evolving well into training, not settled at a constant floor.

6.6 The entropy-related bias, and what remains open

SAC’s maximum-entropy objective introduces a real bias independent of all of the above: the auto-tuned temperature α_{SAC} is adjusted to hold the policy’s entropy at `target_entropy`, fixed at 1.0 for the entire run in `config_sac_learner.py` and never annealed in the code this document currently describes. The soft-optimal policy therefore maximizes $\mathbb{E}[Q(s, a)] + \alpha_{\text{SAC}} \mathcal{H}(\pi(\cdot | s))$, not $\mathbb{E}[Q(s, a)]$ alone, and the gap from the true argmax is bounded by a term of order α_{SAC} that Section 6.5 shows is not shrinking. Annealing `target_entropy` (or decaying α_{SAC}) is a plausible, minimally-invasive fix — a training schedule change, not an architectural one — but whether it closes this gap in practice, and whether closing it changes the trained policy’s realized $\text{EE}(\mathbf{W})$, are both untested: an earlier pilot run with annealing enabled was launched but its outcome was never checked before the code was reverted to the non-annealed version described here.

A related, purely empirical stabilization idea is worth noting separately: the λ update in (8) is, mathematically, a pure integral controller on the rate/power gap, known from control theory to be prone to oscillation and overshoot. PID-Lagrangian methods from safe RL [23] add proportional and derivative terms to exactly this kind of update to damp such oscillation. This is a cheap engineering fix for training stability; it comes with no convergence proof of its own.

What would be needed for a genuine rigor result, beyond what is established here: either (i) a local, KKT-style optimality characterization at a joint stationary point (weaker than global optimality but provable, and a more promising target than what was attempted before), or (ii) an algorithmic change — replacing the fixed-window EMA with a diminishing-step-size update for λ — that would make the asymptotic separation classical two-timescale theory needs actually hold, at the cost of λ becoming less responsive late in training. The corrected reading of [18, 19] in Section 6 above is the more promising literature home for either direction than the tabular result this section originally relied on. Neither is carried out here; both are left as concrete, well-specified future work rather than as claims this document makes.

7 Related work

The reference paper [1] itself continues an earlier single-objective (sum-rate only) line of work by the same group, starting with the first application of SAC to this exact robust-precoding problem [3] and its extension to a flexible multi-scenario framework [2], which together introduced the SAC-based precoding framework this note builds on. The same group has also directly compared RSMA against SDMA and OMA precoding on this system under CSIT error [4], though without an energy-efficiency objective — relevant given the RSMA framing of the codebase this note documents, but a separate multiple-access question from the EE one this note addresses. Outside that lineage, three adjacent bodies of work are relevant, and it is worth being precise about what each does and does not share with the extension proposed here.

Energy-efficient beamforming via deep RL. DRL has been applied directly to EE-objective beamforming design, e.g. uplink beamforming for long-term EE maximization in cell-free networks via DDPG [9]. This confirms “RL for an EE objective” is an established combination in general, but not in a satellite downlink setting and not paired with CSIT-error robustness.

Energy-efficient learning for LEO satellites. DRL has also been used for EE optimization specifically in LEO satellite systems, e.g. joint beamforming, UAV positioning, and rate-splitting for RSMA-based LEO links assisted by an active reconfigurable intelligent surface [10], and a risk-aware (CVaR-based) DRL framework for robust hybrid beamforming in massive-MIMO LEO systems that also targets energy efficiency [11]. The latter extends an earlier, non-learned CVaR-based robust beamforming formulation by an overlapping author group [12], which solves the same risk-aware hybrid-precoding problem via alternating optimization (no DRL) and targets data-rate QoS under worst-case channel variation rather than an explicit energy-efficiency objective. Together, these three combine EE, robustness, and LEO satellites as we do, but in materially larger/different system architectures (RIS- and UAV-assisted, or massive-MIMO hybrid precoding) and via a different robustness mechanism (risk measures rather than training under a CSIT error distribution).

Fractional programming combined with DRL. The Dinkelbach transform’s classical, non-learned use in beamforming is developed comprehensively in [8]. Closer to our combination, [13] jointly uses fractional programming and DRL for robust EE beamforming in a WPT-assisted D2D/MISO-NOMA setting; there, fractional programming and DRL solve separate subproblems in an alternating scheme, rather than the Dinkelbach multiplier being folded directly into a single RL reward signal as in (8).

Closest classical (non-learned) analogue. [14] solves an EE precoding problem for a multi-gateway, bent-pipe SATCOM system using Dinkelbach’s method combined with an MMSE-based (WMMSE-style) transformation and a compressed-sensing relaxation for sparse feeder-link-beam matching – the same general tool (Dinkelbach) applied to a materially different, and more classically tractable, version of the problem. Three differences separate it from this work: (i) it is a model-based iterative convex optimization, solved assuming accurate channel knowledge at every step, not a learned policy; (ii) it does not train against or evaluate robustness to CSIT *estimation error* – channel impairments (phase noise, fading) are simulated, but the algorithm optimizes assuming it knows the channel it is given; and (iii) Dinkelbach’s λ is updated via the classical exact alternating scheme (solve the inner QCQP to convergence, then update λ), not the online, continuously-updated reward-embedded form of (8). This sharpens the gap this work fills: robustness to erroneous CSIT is difficult to fold into a closed-form/convex-relaxation solver of this kind, which is precisely why [1] argues for a learned policy in the first place – an argument this work extends from the sum-rate objective to the EE one.

None of these combine, on the exact system and CSIT-error-robustness setup of [1], (i) the electrical power model (4), (ii) the resulting EE objective (5), and (iii) an online, reward-embedded Dinkelbach update (8). That combination, on this system, is the specific gap this work fills.

8 Summary: what is the paper’s, what is ours

Component	Status
Channel model, LoS geometry, CSIT error sources (Eq. 1 and §2)	Reproduced unchanged from [1].
Sum-rate objective (2) and always-saturate precoder (3)	Reproduced unchanged; retained as Scheme I, a same-conditions reference point.
SAC algorithm (actor/critic structure, entropy objective, offline replay buffer)	Reused, same underlying learning algorithm.
Electrical power model P_{total} (4) (amplifier efficiency, static circuit power)	New. No analogue in the paper.
Energy-efficiency objective (5) (rate over total power)	New. Paper only ever maximizes rate subject to a power cap.
Clip-only precoder (7) realizing the true inequality constraint	New. Every paper precoder instead implements (3).
Dinkelbach reward reformulation (8) with an online, windowed λ update	New. Not a fractional program in the paper at all.
Honest theoretical accounting for the online λ update (Section 6)	New. Establishes what genuinely holds (bandit reduction, strengthened by $\gamma = 0$; smoothness of the fixed- λ surface; a self-consistency-not-optimality fixed-point argument) and corrects an earlier draft’s overclaimed convergence argument, which misapplied a tabular-only theorem and mischaracterized two other papers’ actual (partially continuous-action) scope. Identifies the entropy-driven bias as real and currently unresolved, not fixed, and states two concrete open directions rather than a closed result.

The claim of novelty rests on Section 4: introducing P_{total} into the objective is what breaks Lemma 1’s always-saturate conclusion, and Section 7 situates that claim against the closest work found elsewhere in the literature. Whether the trained policy *empirically* finds and exploits an interior optimum that beats Scheme I on realized EE($\mathbf{W}$) is a separate, still-open empirical question, addressed by evaluation rather than by the formulation itself. Section 6 addresses a different open question — not whether the trained policy performs well, but whether the training procedure itself has a defensible theoretical basis — and, on correction, establishes less than an earlier draft claimed: self-consistency of the (λ, θ) fixed point, not its optimality, plus two concrete, not-yet-implemented directions (entropy-target annealing; a diminishing-step-size λ update) that remain untested rather than a closed result.

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